

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-01.jpg":

$$L = \lim_{n \rightarrow \infty} e^{-n} \sum_{k=0}^n \frac{n^k}{k!}$$

Detailed Solution

To evaluate the limit L , we can elegantly solve this by interpreting the expression using probability theory, specifically relying on the Poisson distribution and the Central Limit Theorem.

Step 1: Relate to the Poisson Distribution

Consider a Poisson random variable X_n with rate parameter $\lambda = n$. The probability mass function (PMF) of X_n is given by:

$$P(X_n = k) = \frac{n^k e^{-n}}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

The expression inside the limit is exactly the sum of these probabilities from $k = 0$ to $k = n$. Thus, it represents the cumulative distribution function (CDF) evaluated at n :

$$e^{-n} \sum_{k=0}^n \frac{n^k}{k!} = \sum_{k=0}^n P(X_n = k) = P(X_n \leq n)$$

So, the limit we want to evaluate becomes:

$$L = \lim_{n \rightarrow \infty} P(X_n \leq n)$$

Step 2: Apply the Central Limit Theorem

A fundamental property of the Poisson distribution is that a Poisson random variable with parameter n (where n is an integer) can be viewed as the sum of n independent and identically distributed (i.i.d.) Poisson random variables, each with parameter 1.

Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. random variables where $Y_i \sim \text{Poisson}(1)$. Then:

$$X_n = \sum_{i=1}^n Y_i \sim \text{Poisson}(n)$$

For a Poisson(1) variable, the expected value is $\mu = 1$ and the variance is $\sigma^2 = 1$. Therefore, for X_n , the mean is $E[X_n] = n$ and the variance is $\text{Var}(X_n) = n$.

By the Central Limit Theorem, as the sample size $n \rightarrow \infty$, the standardized form of X_n converges in distribution to a standard normal random variable $Z \sim \mathcal{N}(0, 1)$:

$$Z_n = \frac{X_n - E[X_n]}{\sqrt{\text{Var}(X_n)}} = \frac{X_n - n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Step 3: Evaluate the Limit

We want to find the limit of $P(X_n \leq n)$. We can rewrite the inequality $X_n \leq n$ by standardizing both sides, just as we did for the random variable itself:

$$P(X_n \leq n) = P\left(\frac{X_n - n}{\sqrt{n}} \leq \frac{n - n}{\sqrt{n}}\right) = P(Z_n \leq 0)$$

Now, apply the limit as $n \rightarrow \infty$:

$$L = \lim_{n \rightarrow \infty} P(X_n \leq n) = \lim_{n \rightarrow \infty} P(Z_n \leq 0)$$

Since Z_n converges to $Z \sim \mathcal{N}(0, 1)$, and the standard normal distribution is continuous and perfectly symmetric around 0:

$$P(Z \leq 0) = \frac{1}{2}$$

Final Answer:

$$\frac{1}{2}$$